MIL-E-46440D (AR)
25 July 1979
SUPERSEDING
MIL-E-46440C (PA)
18 March 1975

MILITARY SPECIFICATION

ELEMENT, IGNITION, ELECTRIC, M18 PARTS

This Specification is approved for use by the US Army Armament Research and Development Command, Department of the Army and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for one type ignition element designated as Element, Electric, M18.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

ARRADCOM

8831029 - Element, Ignition, Electric, M18 Metal Parts Assembly

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, New York 10018).

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The assembly shall comply with all requirements specified on Drawing (dwg.) 8831029, and with all requirements specified in applicable specifications.

3.3 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship molded parts.-All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be free of burrs, chips, sharp edges, cracks, crazes, unblended radii, porosity, warpeage, burn marks, checks, chipped edges, blisters, excess flash, dirt, grease, rust, salt deposits, visible raised cement seams, solder splash, corrosion products, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part or assembly nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

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3.5 Workmanship metal parts.-All parts and assemblies shall be fabricated and finished in the thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item.

a. First article inspection
b. Quality conformance inspection

4.3 First article inspection

4.3.1 First article inspection.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities indicated.

Part Description	Drawing	Quantity
M18 Ignition Element	8831029	25
Ignition Cup	8831030	25
Electrode	8831031	25
Insulator	8831032	See Note 1, Table I
Insulator Disc	8831033	See Note 1, Table I

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46440D(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Element, Ignition, Electric, M18 Parts				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Element, Ignition, Electric, M18 (Dwg. 8831029)				
	Examination for defects	25		3.2	4.4.2.6
	Electrical resistance check	25		3.2	4.5.1
	Cup, Ignition (Dwg. 8831030)				
	Examination for defects	25		3.1	4.4.2.2
	Electrode (Dwg. 8831031)				
	Examination for defects	25		3.1	4.4.2.3
	Insulator (Dwg. 8831032)	Note 1			
	Examination for defects			3.1	4.4.2.4
	Disc, Insulator (Dwg. 8831033)	Note 1			
	Examination for defects			3.1	4.4.2.5

NOTES: 1. Submit 5 samples from each mold cavity to be used in regular production.

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4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling Plans.-Unless otherwise specified in the Classification of Defects and Tests tables, sampling plans and procedures shall be in accordance with MIL-STD-105, Inspection Level II

4.4.2.1 Visual inspections.-In the event of a dispute, four power (4X) magnification shall be used in performing visual inspections for such defects as burrs and foreign matter. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46440D (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Cup, Ignition		8831030
			NEXT HIGHER ASSEMBLY
			8831029

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	True position of outside diameter with large inside diameter		0.65%	3.1	Gage
202	Length along small inside diameter		0.65%	3.1	Gage
203	Outside diameter, min.		0.65%	3.1	Gage
204	Small inside diameter, min.		0.65%	3.1	Gage
205	Large inside diameter, max.		0.65%	3.1	Gage
206	Total length		0.65%	3.1	Gage
207	Radius missing or incorrect		0.65%	3.1	Gage
208	Chamfer missing or incorrect		0.65%	3.1	Gage
209	Surface finish improper		0.65%	3.1	Gage
210	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46440D (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Electrode		8831031
			NEXT HIGHER ASSEMBLY
			8831029

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR DOCS	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	True position of shoulder diameter with diameter of large end		0.65%	3.1	Gage
202	Length of large end		0.65%	3.1	Gage
203	Length, including shoulder, from large end		0.65%	3.1	Gage
204	Total length		0.65%	3.1	Gage
205	Diameter of shoulder		0.65%	3.1	Gage
206	Diameter of large end		0.65%	3.1	Gage
207	Diameter of small end		0.65%	3.1	Gage
208	Radii missing or incorrect		0.65%	3.1	Gage
209	Chamfer missing or incorrect		0.65%	3.1	Gage
210	Surface finish improper		0.65%	3.1	Gage
211	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46440D (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Insulator (See Note)		8831032
			NEXT HIGHER ASSEMBLY
			8831029

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR DOSS	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	True position of small inside diameter with large inside diameter		0.65%	3.1	Gage
202	Small outside diameter, max.		0.65%	3.1	Gage
203	Distance from small end to shoulder		0.65%	3.1	Gage
204	Large outside diameter, max.		0.65%	3.1	Gage
205	Wall thickness at shoulder		0.65%	3.1	Gage
206	Total length		0.65%	3.1	Gage
207	Large inside diameter, max.		0.65%	3.1	Gage
208	Radii missing or incorrect		0.65%	3.1	Gage
209	Chamfer missing or incorrect		0.65%	3.1	Gage
210	Evidence of poor workmanship, including presence of flash		0.65%	3.4	Visual/Manual

NOTE: 1. See Paragraph 4.4.3.1 for optional sampling plan.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46440D (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Disc, Insulator (See Note)		8831033
			NEXT HIGHER ASSEMBLY 8831029

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Outside diameter, min		0.650	3.1	Gage
202	Inside diameter, max.		0.650	3.1	Gage
203	Thickness		0.650	3.1	Gage
204	Evidence of poor workmanship		0.650	3.4	Visual

NOTES: 1. See paragraph 4.4.3.1 for optional sampling plan.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46440D (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Assembly		8831029
			NEXT HIGHER ASSEMBLY
			8840277

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR LOSS	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bridge wire inadequately swaged as determined with 4X to 6X power magnification		0.40%	3.1	Visual
102	Any component missing		0.40%	3.1	Visual
103	Parts not firmly held together (See Note 1)		0.40%	3.1	Manual
104	Electrical resistance check		100%	3.1	Test/4.5.1
105	Cut off nib on electrode not removed by machining		0.40%	3.1	Visual
Minor					
201	Total length, max.		0.65%	3.1	Gage
202	Outside diameter of cup, max.		0.65%	3.1	Gage
203	Bridge wire extends beyond cylindrical surface of cup		0.65%	3.1	Visual/Manual
204	Insulator cracked		0.65%	3.1	Visual
205	Width of slot in cup, max.		0.65%	3.1	Gage
206	Depth of slot in cup, max.		0.65%	3.1	Gage
207	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:
1. There shall be no evidence of looseness when the ignition cup is held and the electrode or insulator is twisted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46440D (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Cartons, Packed, Prior to Sealing		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Any support missing or damaged		0.65%	3.1	Visual
202	Separator(s) missing or damaged		0.65%	3.1	Visual
203	Improperly packed		0.65%	3.1	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46440D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Cartons, Packed, After Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Contents loose		0.65%	3.1	Manual
202	Carton damaged, cut or gouged through all layers		0.65%	3.1	Visual
203	Carton not properly sealed		0.65%	3.1	Visual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Dimensional control of molded parts.-In place of the normal sampling associated with the classification of defects, and subject to the approval of the contracting officer, the optional sampling plan in paragraph 4.4.3.2 may be used for dimensional control of molded parts. Use of this optional plan is limited to those dimensions and features controlled by the mold cavity. For all specified tests, visual examinations, and examinations of dimensions or features not developed exclusively by the mold cavity, these optional plans are not applicable. In the event of in-process rejects during optional sampling, corrective action applicable to parts produced prior to those rejected, and subsequent to the last acceptable sample, shall require approval of the Contracting Officer.

4.4.3.2 Sampling plan.-A sample of five parts (as molded) shall be inspected to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be dimensionally inspected after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part will be removed from production. Further, that portion of production since the last control check shall be rejected, pending corrective action. All cavities removed from production because of some fault, may, after reworking, be returned to production, providing they pass the qualification test above.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.2. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see 6.2 and Section 6 of MIL-A-48078).

4.5 Test methods and procedures

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4.5.1 Electrical resistance.-The igniter element assembly shall be tested for electrical resistance by applying an electrical current of 20 milliamperes max., using the ignition cup as one terminal and the electrode as the other terminal. Observation shall be made for compliance with the applicable requirement.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing

5.1.1 Level B.-The assembly shall be packaged and packed as specified in MIL-STD-1169.

5.2 Marking.-Marking shall be in accordance with MIL-STD-1169, requirements.

6. NOTES

6.1 Ordering data.-See MIL-A-48078.

6.2 Submission of designs for approval.-Submit equipment designs as required to: Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801 (see 6.2.3 of MIL-A-48078).

6.3 Delivery of data cards to the technical activity.-The following address is to be added to those listed in the contract (Form DD 1423) for delivery of one copy of each Ammunition Data Card prepared by the contractor for the duration of the contract: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-E, Dover, New Jersey 07801.

6.4 Changes from previous issue.-Due to the extensiveness of the changes, asterisks are not used in this revision to identify changes with respect to the previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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